THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of the first 8 terms of the AP $50, 45, 40, 35, \ldots$ is: 1

(a) 260 **(b)** 270 **(c)** 268 **(d)** 252

2. The 4th term of the GP $1, 5, 25, \ldots$ is: 1

(a) 125 **(b)** 124 **(c)** 130 **(d)** 126

3. The number of terms in the AP $100, 94, 88, 82, \ldots$ ending at -2 is: 1

(a) 17 **(b)** 19 **(c)** 20 **(d)** 18

4. In the Fibonacci-type rule $t_n = t_{n-1} + t_{n-2}$, each term is: 1

(a) the sum of the two terms before it **(b)** twice the previous term

(c) the product of the two terms before it **(d)** the difference of the previous two terms

5. If $t_n = 5n^2 - 2$, then t_3 is: 1

(a) 45 **(b)** 42 **(c)** 43 **(d)** 44

6. Which term of the sequence $t_n = 7n - 1$ is 209? 1

(a) 31th term **(b)** 30th term **(c)** 32th term **(d)** 29th term

7. A sequence has $t_1 = 1$ and $t_{n+1} = t_n + 7$. Then t_6 equals: 1

(a) 38 **(b)** 36 **(c)** 35 **(d)** 37

8. The 9th term of the AP $11, 20, 29, 38, \ldots$ is: 1

(a) 84 **(b)** 82 **(c)** 85 **(d)** 83

9. The sum of the first 16 terms of the AP $2, 7, 12, 17, \ldots$ is: 1

(a) 648 **(b)** 632 **(c)** 642 **(d)** 616

10. The 6th term of the GP $5, 10, 20, \ldots$ is: 1

(a) 155 **(b)** 160 **(c)** 161 **(d)** 165

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the first five terms of the sequence with $t_1 = 3$ and $t_{n+1} = 2t_n - 1$. 2

12. A sequence is defined by $t_n = 2n + 9$. Find t_{10} and decide whether 31 is a term. 2

13. Show that $-3, 3, 9, 15, \ldots$ is an AP and find its 11th term. 2

14. Write the first five terms of the sequence with $t_1 = 1$ and $t_{n+1} = 2t_n + 5$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find which term of the AP $11, 23, 35, 47, \ldots$ equals 203. 3

16. Find the sum of the first 25 terms of the AP $1, 4, 7, 10, \ldots$. 3

17. Find the common ratio of the GP $4, 8, 16, \ldots$ and its 9th term. 3

OR

Find which term of the AP $3, 13, 23, 33, \ldots$ equals 393. 3

18. Find the sum of the first 14 terms of the AP $13, 17, 21, 25, \ldots$. 3

OR

Find the common ratio of the GP $5, 15, 45, \ldots$ and its 5th term. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A taxi charges a fixed ₹80 and ₹18 per km. 5

(i) Show that the fares for 1, 2, 3 km form an AP.

(ii) Find the fare for 22 km.

OR

Rohan saves ₹400 in the first month and ₹25 more each month than in the previous month. 5

(i) How much does he save in month 14?

(ii) How much does he save in 14 months in all?

20. A taxi charges a fixed ₹120 and ₹30 per km. 5

(i) Show that the fares for 1, 2, 3 km form an AP.

(ii) Find the fare for 15 km.

OR

Rohan saves ₹50 in the first month and ₹5 more each month than in the previous month. 5

(i) How much does he save in month 24?

(ii) How much does he save in 24 months in all?

END OF QUESTION PAPER — Total: 40 Marks

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